

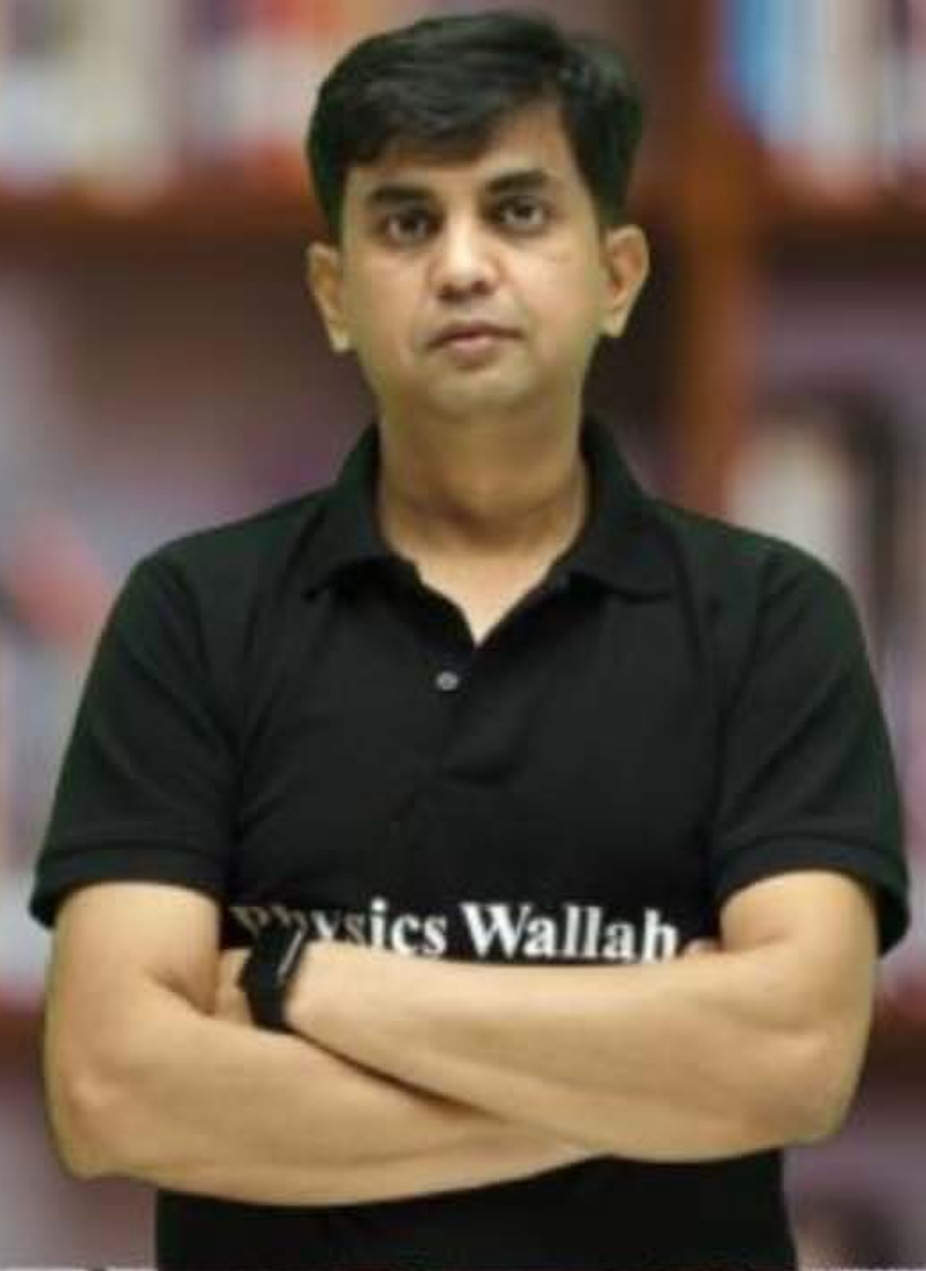
CS & IT ENGINEERING



Computer Network

MAC Sub-Layer

Lecture No. - 01



By - Abhishek Sir



Recap of Previous Lecture



Frame Synchronization

Topic

Byte Count

Topic

Byte Stuffing

Topic

Bit Stuffing





Topics to be Covered



Topic

MAC Sub-Layer

ABOUT ME



Hello, I'm **Abhishek**

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

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#Q. A bit-stuffing based framing protocol uses an 8-bit delimiter pattern of 01111110. If the output bit-string after stuffing is 01111100101, then the input bit-string is stuff bit pattern $\rightarrow$ 01111110

[GATE-2014]

Transmitted code =

- ~~(A) 0111110100~~
- ✓ (B) 0111110101
- ~~(C) 0111111101~~
- ~~(D) 0111111111~~

01111110 01111001010 01111110
 Start Flag 1 2 3 4 5 End Flag
 ↓
 stuff bit

Data = 0111110101 | 01111110
 Frame

Ans: B

#Q. In a data link protocol, the frame delimiter flag is given by 0111. Assuming that bit stuffing is employed, the transmitter sends the data sequence 01110110 as

stuff bit pattern $\rightarrow$ 01110

[GATE-2004]

Data = 01110110
_{1 2 1 2}

Transmitted code =

01110 0110101100 01110

- ~~(A) 01101011~~
- ~~(B) 011010110~~
- ~~(C) 011101100~~
- ✓ (D) 0110101100

Ans: D



Topic : Syllabus



Concept of layering : OSI and TCP/IP Protocol Stacks;

Basics of packet, circuit and virtual circuit switching;

Data link layer : framing, error detection, Medium Access Control, Ethernet bridging; FLOW CONTROL

Routing protocols : shortest path, flooding, distance vector and link state routing;

Fragmentation and IP addressing, IPv4, CIDR notation, Basics of IP support protocols (ARP, DHCP, ICMP), Network Address Translation (NAT);



Topic : Line Configuration

→ Attachment of communication devices to a link.

1. Point-to-Point

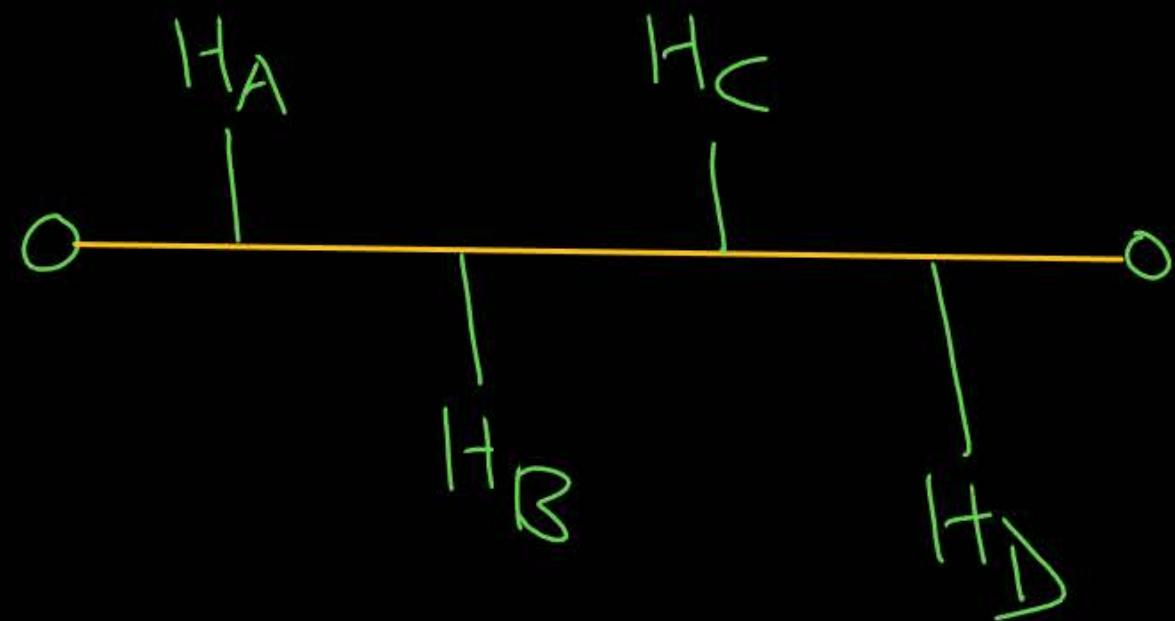
→ Dedicated link between two device
[One sender and one receiver]

2. Multipoint (Multidrop)

→ More than two devices share a single link

→ Broadcast medium
[One sender and all are receiver]

→ e.g. Bus topology, Wireless



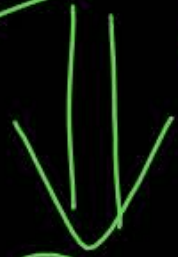


Topic : Data Link Layer



Data link layer services :

1. Framing
2. Error Control
3. Flow Control : [Point to Point Communication]
4. Access Control : Multipoint communication



- (a) Bus Topology
- (b) Ring Topology
- (c) WiFi (Wireless)



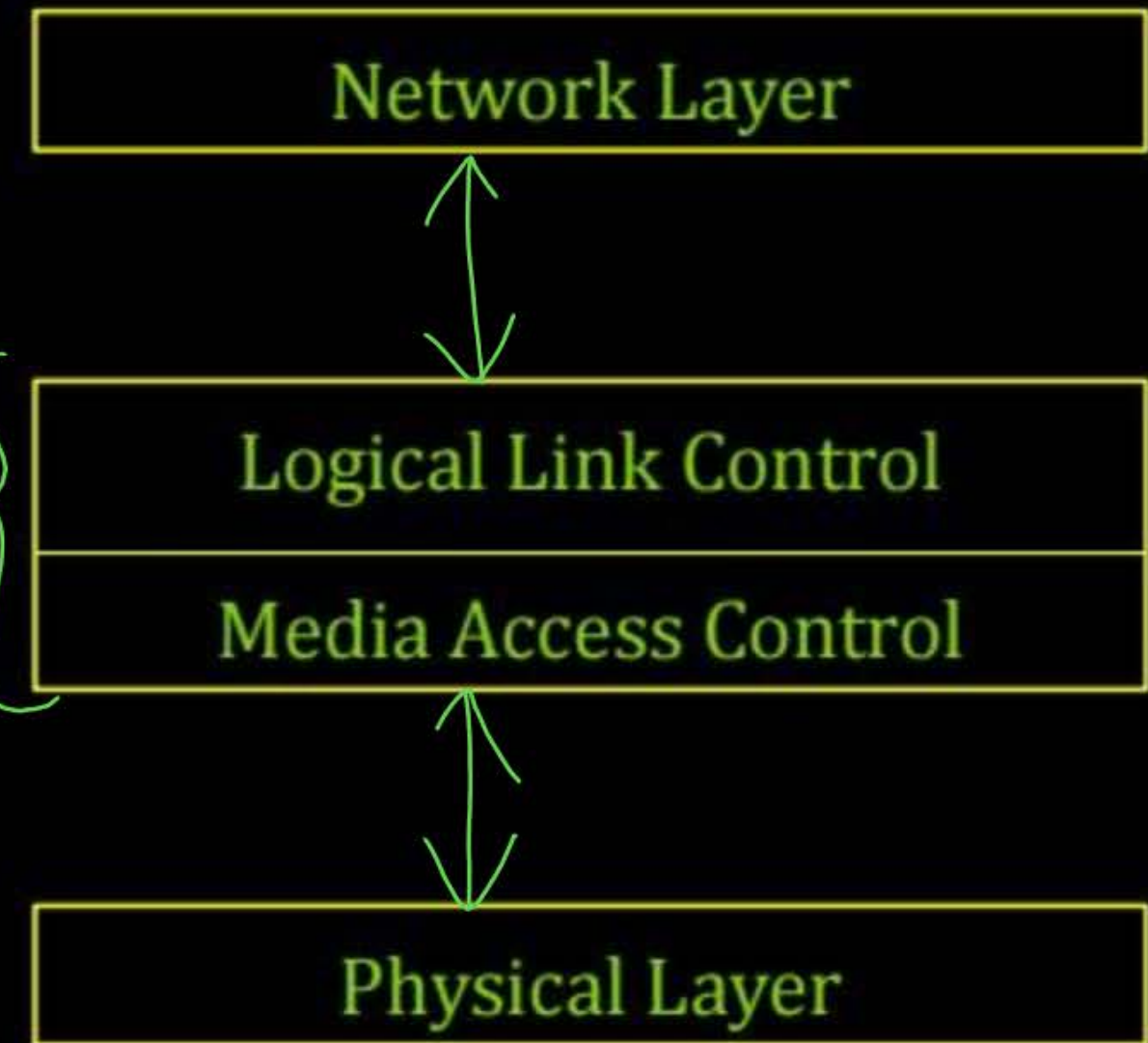
Topic : Data Link Layer



Data link layer is divided into two sub layers :

1. Logical link control (LLC)
2. Media access control (MAC)

Data Link Layer





Topic : Logical Link Control



- > The upper sub layer
- > Responsible for data link control
- > Flow and Error control



Topic : Media Access Control

- > The lower sub layer
- > Determine “who can access the media”
[in multipoint (multidrop) line configuration]
- > No any use, in point-to-point line configuration
- > Multiple access protocols



Topic : MAC Address



- > Identification of Source and Destination nodes of a packet
- > Physical address
[NIC : Network Interface Card]
- > 48 bits or 6 bytes
- > Should be unique

$\Rightarrow$ IP Add. = logical Add.



Topic : MAC Address

-> Two representation of MAC address :

1. Binary Representation [48-bits]

2. Hexadecimal Representation

e.g. "ac:d1:b8:d5:59:0d"

-> Broadcast MAC address (Special MAC address) :

"ff:ff:ff:ff:ff:ff" [1111.....1111]

48 bits



Topic : Multiple Access Protocol

Interference :

-> Two or more simultaneous transmission by nodes

Collision :

-> if node receives two or more signals at same time



Topic : Multiple Access Protocol

Distributed algorithm :

- > Determine "how node shares channel"
- > Determine "when node can transmit"
- > Communication about channel sharing must use channel itself
- > Fully decentralized : No any special node to coordinate



Topic : Multiple Access Protocol

MAC protocols are divided into three categories :

1. Channel Partitioning : Static Allocation
2. Random Access : No any Allocation ✓
3. Taking Turns : Dynamic Allocation



Topic : Channel Partitioning



-> Channelization protocols

-> Divide channel into (smaller pieces)
[Time slot, Frequency or Code]

-> Allocate piece to node for exclusive use

[Static Allocated]

⇒ Best in case of high traffic

⇒ Leads poor utilization of network resources in case of less traffic

⇒ No any chance of collision



Topic : Channel Partitioning



Channel partitioning MAC protocols are :

1. TDMA : [Time-Division Multiple Access]
2. FDMA : [Frequency-Division Multiple Access] $\Rightarrow$ [Ex : Radio FM]
3. CDMA : Code-Division Multiple Access



Topic : Random Access



-> Channel is not divided
[unlike channel partitioning]

* No any allocation
→ chance of collision

-> When node has packet to transmit

it allowed to transmit without any
static or dynamic allocation

⇒ Best when less traffic in the channel.

⇒ Lead more collisions in case of high traffic



Topic : Random Access

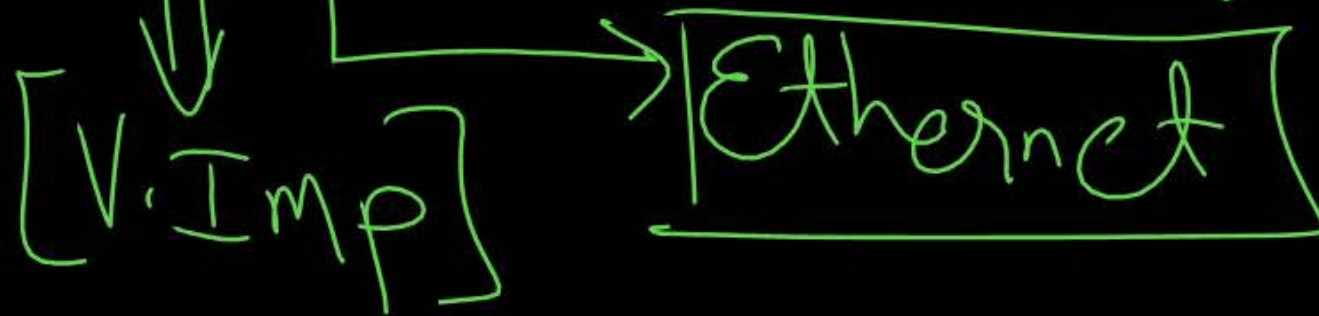
-> Random Access MAC protocols are :

1. **ALOHA** ↓

-> **Pure ALOHA** and **Slotted ALOHA**

2. **CSMA**: Carrier Sense Multiple Access

-> **CSMA/CD** [for wired] and **CSMA/CA** [for wireless]





Topic : Taking Turns



Channel partitioning

- > Static allocation
- > [No chance of collision]
- > Efficient at high traffic/load
[Inefficient at low traffic/load]

Random Access

- > No any allocation
- > [Chance of collision]
- > Efficient at low traffic/load
[Inefficient at high traffic/load]



Topic : Taking Turns



-> Controlled Access Protocol

-> Channel allocated explicitly
[No chance of "collision"]

-> Dynamic allocation



Topic : Taking Turns



-> Controlled Access MAC protocols are :

1. Reservation

2. Polling

3. Token Passing

⇒ Token Bus and Token Ring



Topic : LAN Standards



- IEEE 802.3 : Ethernet ✓ [Bus Topology => CSMA/CD]
- IEEE 802.4 : Token Bus ✗ [Bus Topology => Token]
- IEEE 802.5 : Token Ring ✗ [Ring Topology => Token]
- IEEE 802.11 : Wireless ✗ [Wi-Fi => CSMA/CA]



2 mins Summary



Topic

~~Byte Count~~

MAC sublayer

Topic

~~Byte Stuffing~~

Multiple Access Protocols

Topic

~~Bit Stuffing~~



THANK - YOU

